Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.5)

Metric

$$ds^2 = -\epsilon n \mathrm{d}x \otimes \mathrm{d}x + m \mathrm{d}x \otimes \mathrm{d}y + m \mathrm{d}y \otimes \mathrm{d}x + l \mathrm{d}y \otimes \mathrm{d}y + e^M \mathrm{d}z \otimes \mathrm{d}z + \epsilon e^M \mathrm{d}t \otimes \mathrm{d}t$$

Coordinates

$$(x^\mu) = (x, y, z, t)$$

Metric Functions

$$m(t, z) = m$$

$$n(t, z) = n$$

$$l(t, z) = l$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{\epsilon}\sqrt{n}\right) dx + \left(\frac{1}{2}\sqrt{2}\sqrt{\frac{m^2}{\epsilon n} + l} - \frac{\sqrt{2}m}{2\sqrt{\epsilon}\sqrt{n}}\right) dy$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{\epsilon}\sqrt{n}\right) dx + \left(-\frac{1}{2}\sqrt{2}\sqrt{\frac{m^2}{\epsilon n} + l} - \frac{\sqrt{2}m}{2\sqrt{\epsilon}\sqrt{n}}\right) dy$$

$$m = \left(\frac{1}{2}\sqrt{2}e^{\left(\frac{1}{2}M\right)}\right) dz + \left(\frac{1}{2}i\sqrt{2}\sqrt{\epsilon}e^{\left(\frac{1}{2}M\right)}\right) dt$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}e^{\left(\frac{1}{2}M\right)}\right) dz + \left(-\frac{1}{2}i\sqrt{2}\sqrt{\epsilon}e^{\left(\frac{1}{2}M\right)}\right) dt$$

Contravariant components

$$l = \left(-\frac{\sqrt{\epsilon}l\sqrt{n}}{\sqrt{2}\epsilon ln + \sqrt{2}m^2} - \frac{m^2}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{n}} + \frac{\sqrt{\epsilon ln + m^2}m}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{|n|}}\right) dx + \left(\frac{\sqrt{\epsilon ln + m^2}\sqrt{\epsilon}n}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{|n|}}\right) dy$$

$$n = \left(-\frac{\sqrt{\epsilon}l\sqrt{n}}{\sqrt{2}\epsilon ln + \sqrt{2}m^2} - \frac{m^2}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{n}} - \frac{\sqrt{\epsilon ln + m^2}m}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{|n|}}\right) dx + \left(-\frac{\sqrt{\epsilon ln + m^2}\sqrt{\epsilon}n}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{|n|}}\right) dy$$

$$m = \left(\frac{1}{2}\sqrt{2}e^{\left(-\frac{1}{2}M\right)}\right) dz + \left(\frac{i\sqrt{2}e^{\left(-\frac{1}{2}M\right)}}{2\sqrt{\epsilon}}\right) dt$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}e^{\left(-\frac{1}{2}M\right)}\right) dz + \left(-\frac{i\sqrt{2}e^{\left(-\frac{1}{2}M\right)}}{2\sqrt{\epsilon}}\right) dt$$

Directional Derivatives

$$D = -\frac{\sqrt{\epsilon}l\sqrt{n}}{\sqrt{2}\epsilon ln + \sqrt{2}m^2} - \frac{m^2}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{n}} + \frac{\sqrt{\epsilon ln + m^2}m}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{|n|}}\,\partial_x + \frac{\sqrt{\epsilon ln + m^2}\sqrt{\epsilon}n}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{|n|}}\,\partial_y$$

$$\Delta = -\frac{\sqrt{\epsilon}l\sqrt{n}}{\sqrt{2}\epsilon ln + \sqrt{2}m^2} - \frac{m^2}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{n}} - \frac{\sqrt{\epsilon ln + m^2}m}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{\epsilon}\sqrt{|n|}}\,\partial_x - \frac{\sqrt{\epsilon ln + m^2}\sqrt{\epsilon}n}{\left(\sqrt{2}\epsilon ln + \sqrt{2}m^2\right)\sqrt{|n|}}\,\partial_y$$

$$\delta = \frac{1}{2}\sqrt{2}e^{\left(-\frac{1}{2}M\right)}\,\partial_z + \frac{i\sqrt{2}e^{\left(-\frac{1}{2}M\right)}}{2\sqrt{\epsilon}}\,\partial_t$$

$$\overline{\delta} = \frac{1}{2}\sqrt{2}e^{\left(-\frac{1}{2}M\right)}\,\partial_z - \frac{i\sqrt{2}e^{\left(-\frac{1}{2}M\right)}}{2\sqrt{\epsilon}}\,\partial_t$$

Spin Coefficients

$\rho = 0$

$\sigma = 0$

$$\kappa = \frac{\sqrt{2}\epsilon n^3 \frac{\partial}{\partial z} l}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} - \frac{\sqrt{2}\epsilon l n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)} + \frac{i\sqrt{2}\sqrt{\epsilon} n^3 \frac{\partial}{\partial t} l}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} - \frac{i\sqrt{2}\sqrt{\epsilon} l n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)}$$
$$+ \frac{\sqrt{2} m n^2 \frac{\partial}{\partial z} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} - \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)} - \frac{\sqrt{2} m^2 n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} + \frac{i\sqrt{2} m n^2 \frac{\partial}{\partial t} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}|n|}$$
$$- \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial z} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{|n|}} - \frac{i\sqrt{2} m^2 \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}} - \frac{i\sqrt{2} m^2 n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}|n|}$$
$$+ \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial z} n}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{|n|}} - \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial t} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}} + \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial t} n}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}}$$

$\mu = 0$

$\lambda = 0$

$$\nu = -\frac{\sqrt{2}\epsilon n^3 \frac{\partial}{\partial z} l}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} + \frac{\sqrt{2}\epsilon l n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)} + \frac{i\sqrt{2}\sqrt{\epsilon} n^3 \frac{\partial}{\partial t} l}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} - \frac{i\sqrt{2}\sqrt{\epsilon} l n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)}$$
$$- \frac{\sqrt{2} m n^2 \frac{\partial}{\partial z} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} + \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)} + \frac{\sqrt{2} m^2 n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} + \frac{i\sqrt{2} m n^2 \frac{\partial}{\partial t} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}|n|}$$
$$- \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial z} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{|n|}} - \frac{i\sqrt{2} m^2 \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}} - \frac{i\sqrt{2} m^2 n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}|n|}$$
$$+ \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial z} n}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{|n|}} + \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial t} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}} - \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial t} n}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}}$$

$$
\begin{aligned}
\alpha = & -\frac{\sqrt{2}\epsilon n^2 \frac{\partial}{\partial z} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} + \frac{\sqrt{2}\epsilon n^3 \frac{\partial}{\partial z} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} + \frac{i\sqrt{2}\sqrt{\epsilon} n^2 \frac{\partial}{\partial t} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} - \frac{i\sqrt{2}\sqrt{\epsilon} n^3 \frac{\partial}{\partial t} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} \\
& - \frac{\sqrt{2} m n \frac{\partial}{\partial z} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} + \frac{\sqrt{2} m n^2 \frac{\partial}{\partial z} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} + \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} - \frac{\sqrt{2} m^2 n \frac{\partial}{\partial z} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} \\
& + \frac{i\sqrt{2} m n \frac{\partial}{\partial t} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}} - \frac{i\sqrt{2} m n^2 \frac{\partial}{\partial t} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}|n|} - \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial z} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{|n|}} - \frac{i\sqrt{2} m^2 \frac{\partial}{\partial t} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}} \\
& + \frac{i\sqrt{2} m^2 n \frac{\partial}{\partial t} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}|n|} + \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{|n|}} + \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial t} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}} - \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}}
\end{aligned}
$$

$$
\begin{aligned}
\beta = & \frac{\sqrt{2}\epsilon n^2 \frac{\partial}{\partial z} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} - \frac{\sqrt{2}\epsilon n^3 \frac{\partial}{\partial z} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} + \frac{i\sqrt{2}\sqrt{\epsilon} n^2 \frac{\partial}{\partial t} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} - \frac{i\sqrt{2}\sqrt{\epsilon} n^3 \frac{\partial}{\partial t} l}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} \\
& + \frac{\sqrt{2} m n \frac{\partial}{\partial z} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} - \frac{\sqrt{2} m n^2 \frac{\partial}{\partial z} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} - \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} + \frac{\sqrt{2} m^2 n \frac{\partial}{\partial z} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} \\
& + \frac{i\sqrt{2} m n \frac{\partial}{\partial t} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}} - \frac{i\sqrt{2} m n^2 \frac{\partial}{\partial t} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}|n|} - \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial z} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{|n|}} - \frac{i\sqrt{2} m^2 \frac{\partial}{\partial t} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}} \\
& + \frac{i\sqrt{2} m^2 n \frac{\partial}{\partial t} n}{16\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}|n|} + \frac{\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{|n|}} - \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} n^{\frac{3}{2}} \frac{\partial}{\partial t} m}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}} + \frac{i\sqrt{2}\sqrt{\epsilon l n + m^2} m \sqrt{n} \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}\sqrt{|n|}}
\end{aligned}
$$

$$
\begin{aligned}
\tau = & -\frac{\sqrt{2}\epsilon n^3 \frac{\partial}{\partial z} l}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} - \frac{\sqrt{2}\epsilon l n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} - \frac{i\sqrt{2}\sqrt{\epsilon} n^3 \frac{\partial}{\partial t} l}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} - \frac{i\sqrt{2}\sqrt{\epsilon} l n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} \\
& - \frac{\sqrt{2} m n^2 \frac{\partial}{\partial z} m}{4\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} - \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)} + \frac{\sqrt{2} m^2 n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)|n|} \\
& - \frac{i\sqrt{2} m n^2 \frac{\partial}{\partial t} m}{4\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}|n|} - \frac{i\sqrt{2} m^2 \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}} + \frac{i\sqrt{2} m^2 n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2} M\right)} l n^2 + e^{\left(\frac{1}{2} M\right)} m^2 n\right)\sqrt{\epsilon}|n|}
\end{aligned}
$$

$$\pi = \frac{\sqrt{2}\epsilon n^3 \frac{\partial}{\partial z} l}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} + \frac{\sqrt{2}\epsilon l n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)} - \frac{i\sqrt{2}\sqrt{\epsilon} n^3 \frac{\partial}{\partial t} l}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} - \frac{i\sqrt{2}\sqrt{\epsilon} l n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)}$$
$$+ \frac{\sqrt{2} m n^2 \frac{\partial}{\partial z} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|} + \frac{\sqrt{2} m^2 \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)} - \frac{\sqrt{2} m^2 n \frac{\partial}{\partial z} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)|n|}$$
$$- \frac{i\sqrt{2} m n^2 \frac{\partial}{\partial t} m}{4\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}|n|} - \frac{i\sqrt{2} m^2 \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}} + \frac{i\sqrt{2} m^2 n \frac{\partial}{\partial t} n}{8\left(\epsilon e^{\left(\frac{1}{2}M\right)} l n^2 + e^{\left(\frac{1}{2}M\right)} m^2 n\right)\sqrt{\epsilon}|n|}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)}$$
$$- \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} + \frac{\epsilon^3 l n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} - \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|}$$
$$- \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} + \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)}$$
$$+ \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} + \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} + \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|}$$
$$- \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} - \frac{\epsilon^2 l n^4 \frac{\partial^2}{(\partial t)^2} l}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)}$$
$$- \frac{\epsilon^2 m^2 n^3 \frac{\partial^2}{(\partial z)^2} l}{16\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} + \frac{\epsilon^2 m n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|}$$
$$+ \frac{\epsilon^2 l n^2 \frac{\partial}{\partial z} m^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} - \frac{\epsilon^2 l n^3 \frac{\partial}{\partial z} m^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|}$$
$$- \frac{\epsilon^2 l m n^2 \frac{\partial^2}{(\partial z)^2} m}{8\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} - \frac{\epsilon^2 l m n^3 \frac{\partial^2}{(\partial z)^2} m}{8\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} + \frac{\epsilon^2 l n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)}$$
$$- \frac{\epsilon^2 l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)|n|} - \frac{\epsilon^2 l^2 n \frac{\partial}{\partial t} n^2}{32\left(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n\right)} - \dots$$

$\Phi_{01} = 0$

$$
\begin{aligned}
\Phi_{02} = {} & \frac{\epsilon^3 n^2 \frac{\partial}{\partial z} l^2}{16\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^3 l n^2 \frac{\partial^2}{(\partial z)^2} l}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^3 l^2 \frac{\partial}{\partial z} n^2}{16\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^3 l^2 n \frac{\partial^2}{(\partial z)^2} n}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} l n^2 \frac{\partial^2}{\partial t \partial z} l}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{i\,\epsilon^{\frac{5}{2}} n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i\,\epsilon^{\frac{5}{2}} l^2 n \frac{\partial^2}{\partial t \partial z} n}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{i\,\epsilon^{\frac{5}{2}} l^2 \frac{\partial}{\partial t} n \frac{\partial}{\partial z} n}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\
& - \frac{\epsilon^2 n^2 \frac{\partial}{\partial t} l^2}{16\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 l n^2 \frac{\partial^2}{(\partial t)^2} l}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 m^2 n \frac{\partial^2}{(\partial z)^2} l}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\
& + \frac{\epsilon^2 m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l n \frac{\partial}{\partial z} m^2}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l m n \frac{\partial^2}{(\partial z)^2} m}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l^2 \frac{\partial}{\partial t} n^2}{16\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\
& + \frac{\epsilon^2 l^2 n \frac{\partial^2}{(\partial t)^2} n}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 m^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{\epsilon^2 l m \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{\epsilon^2 l m^2 \frac{\partial^2}{(\partial z)^2} n}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\
& - \frac{i\,\epsilon^{\frac{3}{2}} m^2 n \frac{\partial^2}{\partial t \partial z} l}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{i\,\epsilon^{\frac{3}{2}} m n \frac{\partial}{\partial z} l \frac{\partial}{\partial t} m}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i\,\epsilon^{\frac{3}{2}} l m n \frac{\partial^2}{\partial t \partial z} m}{2\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{i\,\epsilon^{\frac{3}{2}} m n \frac{\partial}{\partial t} l \frac{\partial}{\partial z} m}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} \\
& - \frac{i\,\epsilon^{\frac{3}{2}} l n \frac{\partial}{\partial t} m \frac{\partial}{\partial z} m}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \frac{i\,\epsilon^{\frac{3}{2}} m^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{8\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} + \frac{i\,\epsilon^{\frac{3}{2}} l m \frac{\partial}{\partial z} m \frac{\partial}{\partial t} n}{4\,(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4)} - \dots
\end{aligned}
$$

$\Phi_{10} = 0$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{3\,\epsilon^3 l n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l n^4 \frac{\partial^2}{(\partial t)^2} l}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^2 m^2 n^3 \frac{\partial^2}{(\partial z)^2} l}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 m n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{3\,\epsilon^2 l n^2 \frac{\partial}{\partial z} m^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l n^3 \frac{\partial}{\partial z} m^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 l m n^2 \frac{\partial^2}{(\partial z)^2} m}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^2 l m n^3 \frac{\partial^2}{(\partial z)^2} m}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{3\,\epsilon^2 l n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^2 l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 l^2 n \frac{\partial}{\partial t} n^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \dots
\end{aligned}
$$

$$\Phi_{12} = 0$$

$$
\begin{aligned}
\Phi_{20} = {} & \frac{\epsilon^3 n^2 \frac{\partial}{\partial z} l^2}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^3 l n^2 \frac{\partial^2}{(\partial z)^2} l}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^3 l^2 \frac{\partial}{\partial z} n^2}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^3 l^2 n \frac{\partial^2}{(\partial z)^2} n}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} l n^2 \frac{\partial^2}{\partial t \partial z} l}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{i\,\epsilon^{\frac{5}{2}} n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{i\,\epsilon^{\frac{5}{2}} l^2 n \frac{\partial^2}{\partial t \partial z} n}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{i\,\epsilon^{\frac{5}{2}} l^2 \frac{\partial}{\partial t} n \frac{\partial}{\partial z} n}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& - \frac{\epsilon^2 n^2 \frac{\partial}{\partial t} l^2}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 l n^2 \frac{\partial^2}{(\partial t)^2} l}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 m^2 n \frac{\partial^2}{(\partial z)^2} l}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{\epsilon^2 m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l n \frac{\partial}{\partial z} m^2}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l m n \frac{\partial^2}{(\partial z)^2} m}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l^2 \frac{\partial}{\partial t} n^2}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{\epsilon^2 l^2 n \frac{\partial^2}{(\partial t)^2} n}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 m^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 l m \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l m^2 \frac{\partial^2}{(\partial z)^2} n}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{i\,\epsilon^{\frac{3}{2}} m^2 n \frac{\partial^2}{\partial t \partial z} l}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{i\,\epsilon^{\frac{3}{2}} m n \frac{\partial}{\partial z} l \frac{\partial}{\partial t} m}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{i\,\epsilon^{\frac{3}{2}} l m n \frac{\partial^2}{\partial t \partial z} m}{2\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{i\,\epsilon^{\frac{3}{2}} m n \frac{\partial}{\partial t} l \frac{\partial}{\partial z} m}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{i\,\epsilon^{\frac{3}{2}} l n \frac{\partial}{\partial t} m \frac{\partial}{\partial z} m}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{i\,\epsilon^{\frac{3}{2}} m^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{8\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{i\,\epsilon^{\frac{3}{2}} l m \frac{\partial}{\partial z} m \frac{\partial}{\partial t} n}{4\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \dots
\end{aligned}
$$

$$\Phi_{21} = 0$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^3 l n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 l n^4 \frac{\partial^2}{(\partial t)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 m^2 n^3 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 m n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{\epsilon^2 l n^2 \frac{\partial}{\partial z} m^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 l n^3 \frac{\partial}{\partial z} m^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{\epsilon^2 l m n^2 \frac{\partial^2}{(\partial z)^2} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 l m n^3 \frac{\partial^2}{(\partial z)^2} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 l n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 l^2 n \frac{\partial}{\partial t} n^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \dots
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & \frac{\epsilon^3 n^2 \frac{\partial}{\partial z} l^2}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^3 l n^2 \frac{\partial^2}{(\partial z)^2} l}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^3 l n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^3 l^2 \frac{\partial}{\partial z} n^2}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& - \frac{\epsilon^3 l^2 n \frac{\partial^2}{(\partial z)^2} n}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 n^2 \frac{\partial}{\partial t} l^2}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l n^2 \frac{\partial^2}{(\partial t)^2} l}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 m^2 n \frac{\partial^2}{(\partial z)^2} l}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{\epsilon^2 m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l n \frac{\partial}{\partial z} m^2}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l m n \frac{\partial^2}{(\partial z)^2} m}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{\epsilon^2 l^2 \frac{\partial}{\partial t} n^2}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 l^2 n \frac{\partial^2}{(\partial t)^2} n}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 m^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 l m \frac{\partial}{\partial z} m \frac{\partial}{\partial z} n}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& - \frac{\epsilon^2 l m^2 \frac{\partial^2}{(\partial z)^2} n}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon m^2 n \frac{\partial^2}{(\partial t)^2} l}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon m n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon l n \frac{\partial}{\partial t} m^2}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& - \frac{\epsilon l m n \frac{\partial^2}{(\partial t)^2} m}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon m^2 \frac{\partial}{\partial z} m^2}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon m^3 \frac{\partial^2}{(\partial z)^2} m}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon m^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{16\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{\epsilon l m \frac{\partial}{\partial t} m \frac{\partial}{\partial t} n}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon l m^2 \frac{\partial^2}{(\partial t)^2} n}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{m^2 \frac{\partial}{\partial t} m^2}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{m^3 \frac{\partial^2}{(\partial t)^2} m}{12\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = {} & \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{i\,\epsilon^{\frac{5}{2}} n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^2 \frac{\partial^2}{\partial t \partial z} n}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^3 \frac{\partial^2}{\partial t \partial z} n}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l^2 n \frac{\partial}{\partial t} n \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^2 \frac{\partial}{\partial t} n \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l n^4 \frac{\partial^2}{(\partial t)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 m^2 n^3 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 m n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \cdots
\end{aligned}
$$

$$\Psi_1 = 0$$

$$
\begin{aligned}
\Psi_2 = {} & \frac{\epsilon^3 n^2 \frac{\partial}{\partial z} l^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} - \frac{\epsilon^3 l n^2 \frac{\partial^2}{(\partial z)^2} l}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} \\
& + \frac{\epsilon^3 l n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^3 l n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} + \frac{\epsilon^3 l^2 \frac{\partial}{\partial z} n^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& + \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} - \frac{\epsilon^3 l^2 n \frac{\partial^2}{(\partial z)^2} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} \\
& + \frac{\epsilon^2 n^2 \frac{\partial}{\partial t} l^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} - \frac{\epsilon^2 l n^2 \frac{\partial^2}{(\partial t)^2} l}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& - \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} - \frac{\epsilon^2 m^2 n \frac{\partial^2}{(\partial z)^2} l}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} \\
& + \frac{\epsilon^2 m n \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} - \frac{\epsilon^2 l m n \frac{\partial^2}{(\partial z)^2} m}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} \\
& - \frac{\epsilon^2 l m n^2 \frac{\partial^2}{(\partial z)^2} m}{24\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} + \frac{\epsilon^2 l n \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 l n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} \\
& + \frac{\epsilon^2 l^2 \frac{\partial}{\partial t} n^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} + \frac{\epsilon^2 l^2 n \frac{\partial}{\partial t} n^2}{96\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)|n|} - \frac{\epsilon^2 l^2 n \frac{\partial^2}{(\partial t)^2} n}{48\left(\epsilon^3 e^M l^2 n^2 + 2\,\epsilon^2 e^M l m^2 n + \epsilon e^M m^4\right)} - \cdots
\end{aligned}
$$

$$\Psi_3 = 0$$

$$
\begin{aligned}
\Psi_4 = {} & \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{i\,\epsilon^{\frac{5}{2}} n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^2 \frac{\partial^2}{\partial t \partial z} n}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^3 \frac{\partial^2}{\partial t \partial z} n}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} l^2 n \frac{\partial}{\partial t} n \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^2 \frac{\partial}{\partial t} n \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{32\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l n^4 \frac{\partial^2}{(\partial t)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{\epsilon^2 m^2 n^3 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 m n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{8\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \dots
\end{aligned}
$$

Newman-Penrose Equations

$$
\begin{aligned}
NP1 = & -\frac{\epsilon^3 \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 n^5 \frac{\partial}{\partial z} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{\epsilon^3 l n^5 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{\epsilon^3 l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 l^2 n^3 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{\epsilon^3 l^2 n^4 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{3 i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{3 i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^2 \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^2 n^5 \frac{\partial}{\partial t} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \dots
\end{aligned}
$$

$$
\begin{aligned}
NP2 = & -\frac{\epsilon^3 \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 n^5 \frac{\partial}{\partial z} l^2}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{\epsilon^3 l n^5 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& - \frac{\epsilon^3 \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{3\,\epsilon^3 l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 l^2 n^3 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{\epsilon^3 l^2 n^4 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} l n^5 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{4\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} n^5 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{4\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{3 i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^3 \frac{\partial^2}{\partial t \partial z} n}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \dots
\end{aligned}
$$

$$
\begin{aligned}
NP6 = {} & \frac{\epsilon^3 n^3 \frac{\partial}{\partial z} l^2}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^3 l n^3 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{17\,\epsilon^3 l n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{17\,\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{5\,\epsilon^3 l^2 n \frac{\partial}{\partial z} n^2}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{5\,\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{5\,\epsilon^3 l^2 n^2 \frac{\partial^2}{(\partial z)^2} n}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& - \frac{5\,\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 n^3 \frac{\partial}{\partial t} l^2}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& - \frac{\epsilon^2 l n^3 \frac{\partial^2}{(\partial t)^2} l}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} + \frac{\epsilon^2 l n^4 \frac{\partial^2}{(\partial t)^2} l}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 m^2 n^2 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^2 m^2 n^3 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{\epsilon^2 m n^2 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{48\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{\epsilon^2 m n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{48\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} \\
& + \frac{7\,\epsilon^2 l n^2 \frac{\partial}{\partial z} m^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{7\,\epsilon^2 l n^3 \frac{\partial}{\partial z} m^2}{64\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \frac{\epsilon^2 l m n^2 \frac{\partial^2}{(\partial z)^2} m}{48\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} \\
& + \frac{\epsilon^2 l m n^3 \frac{\partial^2}{(\partial z)^2} m}{48\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} + \frac{17\,\epsilon^2 l n^2 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)} - \frac{17\,\epsilon^2 l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial t} n}{192\,(\epsilon^3 e^M l^2 n^3 + 2\,\epsilon^2 e^M l m^2 n^2 + \epsilon e^M m^4 n)|n|} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
NP7 = {} & -\frac{\epsilon^3 |n| \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} - \frac{5\,\epsilon^3 |n| n^5 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{5\,\epsilon^3 n^6 \frac{\partial}{\partial z} l^2}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} \\
& + \frac{\epsilon^3 |n| l n^5 \frac{\partial^2}{(\partial z)^2} l}{8\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} - \frac{\epsilon^3 l n^6 \frac{\partial^2}{(\partial z)^2} l}{8\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} - \frac{\epsilon^3 |n| \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} \\
& + \frac{5\,\epsilon^3 |n| l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} - \frac{\epsilon^3 l n^5 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{i\,\epsilon^{\frac{5}{2}} |n| \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} |n| l n^5 \frac{\partial^2}{\partial t \partial z} l}{4\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{i\,\epsilon^{\frac{5}{2}} l n^6 \frac{\partial^2}{\partial t \partial z} l}{4\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{i\,\epsilon^{\frac{5}{2}} |n| \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} \\
& + \frac{5 i\,\epsilon^{\frac{5}{2}} |n| n^5 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} - \frac{5 i\,\epsilon^{\frac{5}{2}} n^6 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{i\,\epsilon^{\frac{5}{2}} |n| \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} \\
& - \frac{5 i\,\epsilon^{\frac{5}{2}} |n| l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{i\,\epsilon^{\frac{5}{2}} l n^5 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{i\,\epsilon^{\frac{5}{2}} |n| \overline{n} l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} \\
& - \frac{5 i\,\epsilon^{\frac{5}{2}} |n| l n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{i\,\epsilon^{\frac{5}{2}} l n^5 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \frac{\epsilon^2 |n| \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^6 + 2\,\epsilon^2 e^M l m^2 n^5 + \epsilon e^M m^4 n^4)} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
NP8 = & -\frac{\epsilon^3 \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{12\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 n^5 \frac{\partial}{\partial z} l^2}{12\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{5\,\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{48\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{5\,\epsilon^3 l n^5 \frac{\partial^2}{(\partial z)^2} l}{48\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& - \frac{\epsilon^3 \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{11\,\epsilon^3 l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{\epsilon^3 l^2 n^3 \frac{\partial}{\partial z} n^2}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{48\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 l^2 n^4 \frac{\partial^2}{(\partial z)^2} n}{48\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{5i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{5i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{\epsilon^2 \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{12\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \dots
\end{aligned}
$$

$$
\begin{aligned}
NP10 = & \frac{\epsilon^3 \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{\epsilon^3 n^5 \frac{\partial}{\partial z} l^2}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 l n^5 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{\epsilon^3 \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{3\,\epsilon^3 l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{\epsilon^3 l^2 n^3 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 l^2 n^4 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{i\,\epsilon^{\frac{5}{2}} l n^5 \frac{\partial^2}{\partial t \partial z} l}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{4\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{i\,\epsilon^{\frac{5}{2}} n^5 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l}{4\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{3i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l^2 n^3 \frac{\partial^2}{\partial t \partial z} n}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \dots
\end{aligned}
$$

$$
\begin{aligned}
NP12 = {} & \frac{\epsilon^3 \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{7\,\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{7\,\epsilon^3 n^5 \frac{\partial}{\partial z} l^2}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{7\,\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{7\,\epsilon^3 l n^5 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{\epsilon^3 \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{25\,\epsilon^3 l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} \\
& + \frac{\epsilon^3 l^2 n^3 \frac{\partial}{\partial z} n^2}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 l^2 n^4 \frac{\partial^2}{(\partial z)^2} n}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{\epsilon^2 \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{7\,\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{7\,\epsilon^2 n^5 \frac{\partial}{\partial t} l^2}{192\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{7\,\epsilon^2 l n^4 \frac{\partial^2}{(\partial t)^2} l}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{7\,\epsilon^2 l n^5 \frac{\partial^2}{(\partial t)^2} l}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^2 \overline{\frac{\partial}{\partial z} n} m^2 n^3 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{7\,\epsilon^2 m^2 n^3 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{7\,\epsilon^2 m^2 n^4 \frac{\partial^2}{(\partial z)^2} l}{96\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^2 \overline{\frac{\partial}{\partial z} n} l m n^3 \frac{\partial}{\partial z} m}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{7\,\epsilon^2 m n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{48\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{7\,\epsilon^2 m n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} m}{48\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \dots
\end{aligned}
$$

$$
\begin{aligned}
NP14 = {} & - \frac{\epsilon^3 \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 n^4 \frac{\partial}{\partial z} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 n^5 \frac{\partial}{\partial z} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{\epsilon^3 l n^4 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{\epsilon^3 l n^5 \frac{\partial^2}{(\partial z)^2} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^3 \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{\epsilon^3 l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial z} n}{8\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{\epsilon^3 l^2 n^2 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^3 l^2 n^3 \frac{\partial}{\partial z} n^2}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{\epsilon^3 l^2 n^3 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{\epsilon^3 l^2 n^4 \frac{\partial^2}{(\partial z)^2} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial z} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{i\,\epsilon^{\frac{5}{2}} \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial z} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} + \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{3 i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial z} l \frac{\partial}{\partial t} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{i\,\epsilon^{\frac{5}{2}} l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} + \frac{i\,\epsilon^{\frac{5}{2}} \overline{n} l n^3 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& - \frac{3 i\,\epsilon^{\frac{5}{2}} l n^4 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} n}{32\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \frac{\epsilon^2 \overline{\frac{\partial}{\partial t} n} l n^4 \frac{\partial}{\partial t} l}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} \\
& + \frac{\epsilon^2 n^4 \frac{\partial}{\partial t} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)} - \frac{\epsilon^2 n^5 \frac{\partial}{\partial t} l^2}{16\,(\epsilon^3 e^M l^2 n^4 + 2\,\epsilon^2 e^M l m^2 n^3 + \epsilon e^M m^4 n^2)|n|} - \dots
\end{aligned}
$$

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\begin{aligned}
NP16 = & -\frac{\epsilon^3|n|\overline{\frac{\partial}{\partial z}n}ln^4\frac{\partial}{\partial z}l}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{5\,\epsilon^3|n|n^5\frac{\partial}{\partial z}l^2}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}+\frac{5\,\epsilon^3n^6\frac{\partial}{\partial z}l^2}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)} \\
& +\frac{\epsilon^3|n|ln^5\frac{\partial^2}{(\partial z)^2}l}{8\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{\epsilon^3ln^6\frac{\partial^2}{(\partial z)^2}l}{8\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{\epsilon^3|n|\overline{n}ln^3\frac{\partial}{\partial z}l\frac{\partial}{\partial z}n}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)} \\
& +\frac{5\,\epsilon^3|n|ln^4\frac{\partial}{\partial z}l\frac{\partial}{\partial z}n}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{\epsilon^3ln^5\frac{\partial}{\partial z}l\frac{\partial}{\partial z}n}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{i\,\epsilon^{\frac{5}{2}}|n|\overline{\frac{\partial}{\partial z}n}ln^4\frac{\partial}{\partial t}l}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)} \\
& +\frac{i\,\epsilon^{\frac{5}{2}}|n|ln^5\frac{\partial^2}{\partial t\partial z}l}{4\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{i\,\epsilon^{\frac{5}{2}}ln^6\frac{\partial^2}{\partial t\partial z}l}{4\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{i\,\epsilon^{\frac{5}{2}}|n|\overline{\frac{\partial}{\partial t}n}ln^4\frac{\partial}{\partial z}l}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)} \\
& -\frac{5i\,\epsilon^{\frac{5}{2}}|n|n^5\frac{\partial}{\partial t}l\frac{\partial}{\partial z}l}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}+\frac{5i\,\epsilon^{\frac{5}{2}}n^6\frac{\partial}{\partial t}l\frac{\partial}{\partial z}l}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{i\,\epsilon^{\frac{5}{2}}|n|\overline{n}ln^3\frac{\partial}{\partial z}l\frac{\partial}{\partial t}n}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)} \\
& +\frac{5i\,\epsilon^{\frac{5}{2}}|n|ln^4\frac{\partial}{\partial z}l\frac{\partial}{\partial t}n}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{i\,\epsilon^{\frac{5}{2}}ln^5\frac{\partial}{\partial z}l\frac{\partial}{\partial t}n}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{i\,\epsilon^{\frac{5}{2}}|n|\overline{n}ln^3\frac{\partial}{\partial t}l\frac{\partial}{\partial z}n}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)} \\
& +\frac{5i\,\epsilon^{\frac{5}{2}}|n|ln^4\frac{\partial}{\partial t}l\frac{\partial}{\partial z}n}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}-\frac{i\,\epsilon^{\frac{5}{2}}ln^5\frac{\partial}{\partial t}l\frac{\partial}{\partial z}n}{32\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}+\frac{\epsilon^2|n|\overline{\frac{\partial}{\partial t}n}ln^4\frac{\partial}{\partial t}l}{16\,(\epsilon^3e^Ml^2n^6+2\,\epsilon^2e^Mlm^2n^5+\epsilon e^Mm^4n^4)}+\ldots
\end{aligned}
$$

$$
\begin{aligned}
NP17 = & \frac{8\epsilon^3n^5(\partial_zl)^2+8\epsilon^2n^5(\partial_tl)^2+32\epsilon^2mn^4\partial_zl\,\partial_zm+32\epsilon mn^4\partial_tl\,\partial_tm}{96(\epsilon^3e^Ml^2n^4+2\epsilon^2e^Mlm^2n^3+\epsilon e^Mm^4n^2)|n|}-\frac{4(3\epsilon ln^4-5m^2n^3)(\partial_tm)^2+4(3\epsilon^2ln^4-5\epsilon m^2n^3)(\partial_zm)^2}{96(\epsilon^3e^Ml^2n^4+2\epsilon^2e^Mlm^2n^3+\epsilon e^Mm^4n^2)|n|} \\
& +\frac{6((\epsilon^2ln^4+\epsilon m^2n^3)\partial_tl+2(\epsilon lmn^3+m^3n^2)\partial_tm-(\epsilon lm^2n^2+m^4n)\partial_tn)\overline{\partial_tn}+6((\epsilon^3ln^4+\epsilon^2m^2n^3)\partial_zl+2(\epsilon^2lmn^3+\epsilon m^3n^2)\partial_zm-(\epsilon^2lm^2n^2+\epsilon m^4n)\partial_zn)\overline{\partial_zn}}{96(\epsilon^3e^Ml^2n^4+2\epsilon^2e^Mlm^2n^3+\epsilon e^Mm^4n^2)|n|} \\
& -\frac{10(\epsilon^2ln^5+\epsilon m^2n^4)\partial_t^2l+10(\epsilon^3ln^5+\epsilon^2m^2n^4)\partial_z^2l+20(\epsilon lmn^4+m^3n^3)\partial_t^2m+20(\epsilon^2lmn^4+\epsilon m^3n^3)\partial_z^2m}{96(\epsilon^3e^Ml^2n^4+2\epsilon^2e^Mlm^2n^3+\epsilon e^Mm^4n^2)|n|} \\
& +\frac{2(\epsilon^2l^2n^4+7\epsilon lm^2n^3+6m^4n^2-(\epsilon^2l^2n^3+7\epsilon lm^2n^2+6m^4n)|n|)\partial_t^2n+2(\epsilon^3l^2n^4+7\epsilon^2lm^2n^3+6\epsilon m^4n^2-(\epsilon^3l^2n^3+7\epsilon^2lm^2n^2+6\epsilon m^4n)|n|)\partial_z^2n}{96(\epsilon^3e^Ml^2n^4+2\epsilon^2e^Mlm^2n^3+\epsilon e^Mm^4n^2)|n|}+\ldots
\end{aligned}
$$

$$NP18 = 0$$

Bianchi Identities

$$
\begin{aligned}
BI1 = {} & \frac{(24i\sqrt{2}\epsilon mn^4\partial_t l(\partial_t m)^2 + (i\sqrt{2}\epsilon ln^4 + 25i\sqrt{2}m^2n^3)(\partial_t m)^3 - 6(i\sqrt{2}\epsilon|n|lm^3 - i\sqrt{2}\epsilon lm^3n)(\partial_t n)^3 - 8(i\sqrt{2}\epsilon^3 ln^5 + i\sqrt{2}\epsilon^2m^2n^4)\partial_z l\,\partial_{tz}^2 m)\sqrt{\epsilon ln + m^2}\sqrt{|n|}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{((24i\sqrt{2}\epsilon^2mn^4\partial_t l + (i\sqrt{2}\epsilon^2ln^4 + 25i\sqrt{2}\epsilon m^2n^3)\partial_t m)(\partial_z m)^2 - 8(-i\sqrt{2}\epsilon^2lmn^4 - i\sqrt{2}\epsilon m^3n^3)\partial_t m\,\partial_z^2 m)\sqrt{\epsilon ln + m^2}\sqrt{|n|}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{((((7i\sqrt{2}\epsilon^2lmn^2 - 5i\sqrt{2}\epsilon m^3n)\partial_t l - 6(-3i\sqrt{2}\epsilon lm^2n + 2i\sqrt{2}m^4)\partial_t m)|n| + (-7i\sqrt{2}\epsilon^2lmn^3 + 5i\sqrt{2}\epsilon m^3n^2)\partial_t l - 6(3i\sqrt{2}\epsilon lm^2n^2 - 2i\sqrt{2}m^4n)\partial_t m)(\partial_t n)^2)\sqrt{\epsilon ln + m^2}\sqrt{|n|}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{(-i\sqrt{2}\epsilon^4n^6\partial_t l(\partial_z l)^2 - i\sqrt{2}\epsilon^3n^6(\partial_t l)^3 - 3(-i\sqrt{2}\epsilon^2ln^5 + 3i\sqrt{2}\epsilon m^2n^4)\partial_t l(\partial_t m)^2 - 2(-3i\sqrt{2}\epsilon lmn^4 + i\sqrt{2}m^3n^3)(\partial_t m)^3 + (-i\sqrt{2}\epsilon^2|n|l^2m^2n + i\sqrt{2}\epsilon^2l^2m^2n^2)(\partial_t n)^3)\sqrt{n}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& - \frac{(2(-i\sqrt{2}\epsilon^3ln^6 - i\sqrt{2}\epsilon^2m^2n^5)\partial_t l\,\partial_t^2 l + 2(-i\sqrt{2}\epsilon^4ln^6 - i\sqrt{2}\epsilon^3m^2n^5)\partial_{tz}^2 l\,\partial_z l + 4(-i\sqrt{2}\epsilon^3lmn^5 - i\sqrt{2}\epsilon^2m^3n^4)\partial_z l\,\partial_{tz}^2 m)\sqrt{n}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{(((11i\sqrt{2}\epsilon^3ln^5 + 7i\sqrt{2}\epsilon^2m^2n^4)\partial_t l - 2(-3i\sqrt{2}\epsilon^2lmn^4 + i\sqrt{2}\epsilon m^3n^3)\partial_t m)(\partial_z m)^2 - 16(-i\sqrt{2}\epsilon^3l^2n^5 - 2i\sqrt{2}\epsilon^2lm^2n^4 - i\sqrt{2}\epsilon m^4n^3)\partial_t m\,\partial_z^2 m)\sqrt{n}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} + \cdots
\end{aligned}
$$

$$BI2 = 0$$

$$BI3 = 0$$

$$
\begin{aligned}
BI4 = {} & \frac{(-24i\sqrt{2}\epsilon mn^4\partial_t l(\partial_t m)^2 + (-i\sqrt{2}\epsilon ln^4 - 25i\sqrt{2}m^2n^3)(\partial_t m)^3 - 6(-i\sqrt{2}\epsilon|n|lm^3 + i\sqrt{2}\epsilon lm^3n)(\partial_t n)^3 - 8(-i\sqrt{2}\epsilon^3 ln^5 - i\sqrt{2}\epsilon^2m^2n^4)\partial_z l\,\partial_{tz}^2 m)\sqrt{\epsilon ln + m^2}\sqrt{|n|}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{((-24i\sqrt{2}\epsilon^2mn^4\partial_t l + (-i\sqrt{2}\epsilon^2ln^4 - 25i\sqrt{2}\epsilon m^2n^3)\partial_t m)(\partial_z m)^2 - 8(i\sqrt{2}\epsilon^2lmn^4 + i\sqrt{2}\epsilon m^3n^3)\partial_t m\,\partial_z^2 m)\sqrt{\epsilon ln + m^2}\sqrt{|n|}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{((((-7i\sqrt{2}\epsilon^2lmn^2 + 5i\sqrt{2}\epsilon m^3n)\partial_t l - 6(3i\sqrt{2}\epsilon lm^2n - 2i\sqrt{2}m^4)\partial_t m)|n| + (7i\sqrt{2}\epsilon^2lmn^3 - 5i\sqrt{2}\epsilon m^3n^2)\partial_t l - 6(-3i\sqrt{2}\epsilon lm^2n^2 + 2i\sqrt{2}m^4n)\partial_t m)(\partial_t n)^2\sqrt{\epsilon ln + m^2}\sqrt{|n|}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{(-i\sqrt{2}\epsilon^4n^6\partial_t l(\partial_z l)^2 - i\sqrt{2}\epsilon^3n^6(\partial_t l)^3 - 3(-i\sqrt{2}\epsilon^2ln^5 + 3i\sqrt{2}\epsilon m^2n^4)\partial_t l(\partial_t m)^2 - 2(-3i\sqrt{2}\epsilon lmn^4 + i\sqrt{2}m^3n^3)(\partial_t m)^3 + (-i\sqrt{2}\epsilon^2|n|l^2m^2n + i\sqrt{2}\epsilon^2l^2m^2n^2)(\partial_t n)^3)\sqrt{n}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& - \frac{(2(-i\sqrt{2}\epsilon^3ln^6 - i\sqrt{2}\epsilon^2m^2n^5)\partial_t l\,\partial_t^2 l + 2(-i\sqrt{2}\epsilon^4ln^6 - i\sqrt{2}\epsilon^3m^2n^5)\partial_{tz}^2 l\,\partial_z l + 4(-i\sqrt{2}\epsilon^3lmn^5 - i\sqrt{2}\epsilon^2m^3n^4)\partial_z l\,\partial_{tz}^2 m)\sqrt{n}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} \\
& + \frac{(((11i\sqrt{2}\epsilon^3ln^5 + 7i\sqrt{2}\epsilon^2m^2n^4)\partial_t l - 2(-3i\sqrt{2}\epsilon^2lmn^4 + i\sqrt{2}\epsilon m^3n^3)\partial_t m)(\partial_z m)^2 - 16(-i\sqrt{2}\epsilon^3l^2n^5 - 2i\sqrt{2}\epsilon^2lm^2n^4 - i\sqrt{2}\epsilon m^4n^3)\partial_t m\,\partial_z^2 m)\sqrt{n}}{128(\epsilon^4e^{3M/2}l^3n^5 + 3\epsilon^3e^{3M/2}l^2m^2n^4 + 3\epsilon^2e^{3M/2}lm^4n^3 + \epsilon e^{3M/2}m^6n^2)\sqrt{\epsilon}|n|\sqrt{n}} + \cdots
\end{aligned}
$$

$$BI5 = 0$$

$$BI6 = 0$$

$$
\begin{aligned}
BI7 = {} & \frac{(i\sqrt{2}\epsilon^4 n^6 \partial_t l(\partial_z l)^2 + i\sqrt{2}\epsilon^3 n^6(\partial_t l)^3 + (i\sqrt{2}\epsilon^2 l n^5 + 13 i\sqrt{2}\epsilon m^2 n^4)\partial_t l(\partial_t m)^2 - 2(-i\sqrt{2}\epsilon l m n^4 - 5i\sqrt{2}m^3 n^3)(\partial_t m)^3)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& + \frac{((2i\sqrt{2}\epsilon^3 l^3 n^3 + 7i\sqrt{2}\epsilon^2 l^2 m^2 n^2 + 6i\sqrt{2}\epsilon l m^4 n + (-4i\sqrt{2}\epsilon^3 l^3 n^2 - 9i\sqrt{2}\epsilon^2 l^2 m^2 n - 6i\sqrt{2}\epsilon l m^4)|n| + (i\sqrt{2}\epsilon^3 l^3 n^2 + i\sqrt{2}\epsilon^2 l^2 m^2 n)\overline{n})(\partial_t n)^3)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& - \frac{(2(i\sqrt{2}\epsilon^3 l n^6 + i\sqrt{2}\epsilon^2 m^2 n^5)\partial_t l\, \partial_t^2 l + 2(-i\sqrt{2}\epsilon^4 l n^6 - i\sqrt{2}\epsilon^3 m^2 n^5)\partial_{tz}^2 l\, \partial_z l + 4(i\sqrt{2}\epsilon^4 l n^6 + i\sqrt{2}\epsilon^3 m^2 n^5)\partial_t l\, \partial_z^2 l + 4(-i\sqrt{2}\epsilon^3 l m n^5 - i\sqrt{2}\epsilon^2 m^3 n^4)\partial_z l\, \partial_{tz}^2 m)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& + \frac{(((-15i\sqrt{2}\epsilon^3 l n^5 - 11i\sqrt{2}\epsilon^2 m^2 n^4)\partial_t l - 2(-i\sqrt{2}\epsilon^2 l m n^4 - 5i\sqrt{2}\epsilon m^3 n^3)\partial_t m)(\partial_z m)^2)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& + \frac{(-i\sqrt{2}\epsilon^4 n^5 \partial_t l(\partial_z l)^2 - i\sqrt{2}\epsilon^3 n^5(\partial_t l)^3 + (-i\sqrt{2}\epsilon^2 l n^4 - 13 i\sqrt{2}\epsilon m^2 n^3)\partial_t l(\partial_t m)^2 - 2(i\sqrt{2}\epsilon l m n^3 + 5i\sqrt{2}m^3 n^2)(\partial_t m)^3)\sqrt{\epsilon}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& - \frac{(2(-i\sqrt{2}\epsilon^3 l n^5 - i\sqrt{2}\epsilon^2 m^2 n^4)\partial_t l\, \partial_t^2 l + 2(i\sqrt{2}\epsilon^4 l n^5 + i\sqrt{2}\epsilon^3 m^2 n^4)\partial_{tz}^2 l\, \partial_z l + 4(-i\sqrt{2}\epsilon^4 l n^5 - i\sqrt{2}\epsilon^3 m^2 n^4)\partial_t l\, \partial_z^2 l + 4(i\sqrt{2}\epsilon^3 l m n^4 + i\sqrt{2}\epsilon^2 m^3 n^3)\partial_z l\, \partial_{tz}^2 m)\sqrt{\epsilon}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& + \frac{(((15i\sqrt{2}\epsilon^3 l n^4 + 11i\sqrt{2}\epsilon^2 m^2 n^3)\partial_t l - 2(i\sqrt{2}\epsilon^2 l m n^3 + 5i\sqrt{2}\epsilon m^3 n^2)\partial_t m)(\partial_z m)^2)\sqrt{\epsilon}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
BI8 = {} & -\frac{(i\sqrt{2}\epsilon^4 n^6 \partial_t l(\partial_z l)^2 + i\sqrt{2}\epsilon^3 n^6(\partial_t l)^3 + (i\sqrt{2}\epsilon^2 l n^5 + 13 i\sqrt{2}\epsilon m^2 n^4)\partial_t l(\partial_t m)^2 - 2(-i\sqrt{2}\epsilon l m n^4 - 5i\sqrt{2}m^3 n^3)(\partial_t m)^3)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& - \frac{((2i\sqrt{2}\epsilon^3 l^3 n^3 + 7i\sqrt{2}\epsilon^2 l^2 m^2 n^2 + 6i\sqrt{2}\epsilon l m^4 n + (-4i\sqrt{2}\epsilon^3 l^3 n^2 - 9i\sqrt{2}\epsilon^2 l^2 m^2 n - 6i\sqrt{2}\epsilon l m^4)|n| + (i\sqrt{2}\epsilon^3 l^3 n^2 + i\sqrt{2}\epsilon^2 l^2 m^2 n)\overline{n})(\partial_t n)^3)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& + \frac{(2(i\sqrt{2}\epsilon^3 l n^6 + i\sqrt{2}\epsilon^2 m^2 n^5)\partial_t l\, \partial_t^2 l + 2(-i\sqrt{2}\epsilon^4 l n^6 - i\sqrt{2}\epsilon^3 m^2 n^5)\partial_{tz}^2 l\, \partial_z l + 4(i\sqrt{2}\epsilon^4 l n^6 + i\sqrt{2}\epsilon^3 m^2 n^5)\partial_t l\, \partial_z^2 l + 4(-i\sqrt{2}\epsilon^3 l m n^5 - i\sqrt{2}\epsilon^2 m^3 n^4)\partial_z l\, \partial_{tz}^2 m)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& - \frac{(((-15i\sqrt{2}\epsilon^3 l n^5 - 11i\sqrt{2}\epsilon^2 m^2 n^4)\partial_t l - 2(-i\sqrt{2}\epsilon^2 l m n^4 - 5i\sqrt{2}\epsilon m^3 n^3)\partial_t m)(\partial_z m)^2)\sqrt{\epsilon l n + m^2}\sqrt{|n|}\sqrt{n}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& - \frac{(-i\sqrt{2}\epsilon^4 n^5 \partial_t l(\partial_z l)^2 - i\sqrt{2}\epsilon^3 n^5(\partial_t l)^3 + (-i\sqrt{2}\epsilon^2 l n^4 - 13 i\sqrt{2}\epsilon m^2 n^3)\partial_t l(\partial_t m)^2 - 2(i\sqrt{2}\epsilon l m n^3 + 5i\sqrt{2}m^3 n^2)(\partial_t m)^3)\sqrt{\epsilon}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& + \frac{(2(-i\sqrt{2}\epsilon^3 l n^5 - i\sqrt{2}\epsilon^2 m^2 n^4)\partial_t l\, \partial_t^2 l + 2(i\sqrt{2}\epsilon^4 l n^5 + i\sqrt{2}\epsilon^3 m^2 n^4)\partial_{tz}^2 l\, \partial_z l + 4(-i\sqrt{2}\epsilon^4 l n^5 - i\sqrt{2}\epsilon^3 m^2 n^4)\partial_t l\, \partial_z^2 l + 4(i\sqrt{2}\epsilon^3 l m n^4 + i\sqrt{2}\epsilon^2 m^3 n^3)\partial_z l\, \partial_{tz}^2 m)\sqrt{\epsilon}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} \\
& - \frac{(((15i\sqrt{2}\epsilon^3 l n^4 + 11i\sqrt{2}\epsilon^2 m^2 n^3)\partial_t l - 2(i\sqrt{2}\epsilon^2 l m n^3 + 5i\sqrt{2}\epsilon m^3 n^2)\partial_t m)(\partial_z m)^2)\sqrt{\epsilon}}{384(\epsilon^4 e^{3M/2} l^3 n^5 + 3\epsilon^3 e^{3M/2} l^2 m^2 n^4 + 3\epsilon^2 e^{3M/2} l m^4 n^3 + \epsilon e^{3M/2} m^6 n^2)\sqrt{\epsilon}|n|} - \cdots
\end{aligned}
$$

$$BI9 = 0$$

$$
\begin{aligned}
BI10 = & -\frac{4i\sqrt{2}\epsilon^4 n^7 \frac{\partial}{\partial t} l \frac{\partial}{\partial z} l^2}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} - \frac{4i\sqrt{2}\epsilon^3 n^7 \frac{\partial}{\partial t} l^3}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& - \frac{2\left(5i\sqrt{2}\epsilon^2 l n^6 + 29i\sqrt{2}\epsilon m^2 n^5\right) \frac{\partial}{\partial t} l \frac{\partial}{\partial t} m^2}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& - \frac{4\left(5i\sqrt{2}\epsilon l m n^5 + 13i\sqrt{2} m^3 n^4\right) \frac{\partial}{\partial t} m^3}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} + \frac{4i\sqrt{2}\epsilon^3 l^3 n^4 \frac{\partial}{\partial z} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& - \frac{4i\sqrt{2}\epsilon^2 l^2 m^2 n^3 \frac{\partial}{\partial z} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} - \frac{12i\sqrt{2}\epsilon l m^4 n^2 \frac{\partial}{\partial z} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& + \frac{\left(-2i\sqrt{2}\epsilon^3 l^3 n^3 + 10i\sqrt{2}\epsilon^2 l^2 m^2 n^2 + 16i\sqrt{2}\epsilon l m^4 n\right) \frac{\partial^2}{(\partial t)^2} n \frac{\partial}{\partial z} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} - \frac{2\left(\sqrt{2}\epsilon^4 l^2 n^6 + 6\sqrt{2}\epsilon^3 l m^2 n^5 + 5\sqrt{2}\epsilon^2 m^4 n^4\right) \frac{\partial^2}{(\partial z)^2} l \frac{\partial^2}{(\partial t)^2} n \frac{\partial}{\partial z} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& - \frac{8\left(3\sqrt{2}\epsilon^2 l^2 m n^5 + 30\sqrt{2}\epsilon l m^4 n^2 + 16\sqrt{2} m^6 n - \left(\sqrt{2}\epsilon^3 l^3 n^3 + 4\sqrt{2}\epsilon^2 l^2 m^2 n^2 + 5\sqrt{2}\epsilon l m^4 n + 2\sqrt{2} m^6\right)\overline{n}\right)|n| \frac{\partial^2}{(\partial t)^2} n \frac{\partial}{\partial z} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& - \frac{2\left(\sqrt{2}\epsilon^4 l^2 n^5 + 6\sqrt{2}\epsilon^3 l m^2 n^4 + 5\sqrt{2}\epsilon^2 m^4 n^3\right)|n| \frac{\partial}{\partial z} l \frac{\partial^2}{(\partial z)^2} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} - \frac{8\left(3\sqrt{2}\epsilon^3 l^2 m n^4 + 8\sqrt{2}\epsilon^2 l m^3 n^3 + 5\sqrt{2}\epsilon m^5 n^2\right)|n| \frac{\partial}{\partial z} m \frac{\partial^2}{(\partial z)^2} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& + \frac{2\left(\sqrt{2}\epsilon^4 l^2 n^6 + 6\sqrt{2}\epsilon^3 l m^2 n^5 + 5\sqrt{2}\epsilon^2 m^4 n^4\right) \frac{\partial}{\partial z} l \frac{\partial^2}{(\partial z)^2} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} + \frac{8\left(3\sqrt{2}\epsilon^3 l^2 m n^5 + 8\sqrt{2}\epsilon^2 l m^3 n^4 + 5\sqrt{2}\epsilon m^5 n^3\right) \frac{\partial}{\partial z} m \frac{\partial^2}{(\partial z)^2} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} \\
& - \frac{4\left(\sqrt{2}\epsilon^4 l^3 n^6 + 4\sqrt{2}\epsilon^3 l^2 m^2 n^5 + 5\sqrt{2}\epsilon^2 l m^4 n^4 + 2\sqrt{2}\epsilon m^6 n^3 - \left(\sqrt{2}\epsilon^4 l^3 n^5 + 4\sqrt{2}\epsilon^3 l^2 m^2 n^4 + 5\sqrt{2}\epsilon^2 l m^4 n^3 + 2\sqrt{2}\epsilon m^6 n^2\right)|n|\right) \frac{\partial^3}{(\partial z)^3} n}{256\left(\epsilon^4 e^{\left(\frac{3}{2}M\right)} l^3 n^7 + 3\,\epsilon^3 e^{\left(\frac{3}{2}M\right)} l^2 m^2 n^6 + 3\,\epsilon^2 e^{\left(\frac{3}{2}M\right)} l m^4 n^5 + \epsilon e^{\left(\frac{3}{2}M\right)} m^6 n^4\right)} + \ldots
\end{aligned}
$$

$BI11 = 0$

Petrov Type

Type "D"